Aviv Shaked

Cybersecurity Developer & Researcher · Haifa, Israel

aquamarined07@gmail.com 054-543-5388 github.com/Patcholie patcholie.github.io linkedin.com/in/aviv-shaked-59a4b7271

SUMMARY

C++20 cybersecurity developer and researcher specializing in network security, binary exploitation, and reverse engineering. Graduate of **Magshimim+**, Israel's elite national cybersecurity track, consistently self-driven beyond the curriculum. Co-developed **HitchHiker**, a production covert communication platform embedding encrypted messages into live traffic with zero new packets; placed **2nd globally** in AI4Y (2023).

TECHNICAL SKILLS

Languages	**C++20** · **Python** · C · x86-64 Assembly · SQL · Bash · C#
C++ & Systems	RAII · smart pointers · multi-threading · socket programming (TCP/UDP/raw) · CMake · modular builds
Security	**Binary exploitation** · ROP chains · heap exploitation · format strings · ASLR bypass · **reverse engineering** · **network steganography** · threat modeling · HackTheBox (RE, Pwn, Web, Crypto, Forensics)
Networking	L2–L7 stack · TCP/IP · DNS · TLS · ICMP · ARP · DHCP · bit-level field manipulation · kernel packet interception (NFQueue) · iptables · Wireshark · Scapy
Cryptography	ECDH · ChaCha20 · AES · Argon2id · HKDF · SHA-256 · key exchange · key derivation · hash attacks · applied in production via Crypto++
AI / ML	Neural networks · reinforcement learning · self-play · PyTorch · TensorFlow · Scikit-learn · LLM/API integration
Tools	**Binary Ninja** · IDA Pro · Ghidra · GDB · Wireshark · nmap · Burp Suite · Docker · Git · Kali Linux · AWS EC2 · JetBrains

PROJECTS

HitchHiker: Network Steganography Platform
C++20 · 2025-2026 · Completed

Co-developed with Daniel Varkin · Source available on request · Architecture: patcholie.github.io

C++20 | Crypto++ | NFQueue | FTXUI | Redis | SQLite | Docker | AWS EC2 | CMake

- Co-architected a covert communication platform that embeds encrypted messages into live network traffic: **zero new packets generated**, no observable network presence, undetectable by design
- Full cryptography stack (key exchange, symmetric encryption, key derivation, password hashing) applied at bit-field precision across multiple protocol layers via kernel-level packet interception
- **Pluggable carrier architecture** behind a unified HitchHikerAPI; three independent CMake targets (lib / server / client), Redis + SQLite storage, full TUI (FTXUI); deployed on AWS EC2 via Docker

Multiplayer Trivia Game System
C++ / C# / Unity · Magshimim · 2024

C++ | C# | Unity | SQLite | TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server, C# Unity client, and SQLite database integrated end-to-end with clean OOP architecture; scored 130/100 including bonus marks

Analisis: AI Wound Treatment Platform
Python · 2023 · Public

- Co-led development of an AI-powered wound diagnosis and treatment guidance platform; **placed 2nd globally** in the AI4Y World Competition (2023) across international teams

Mancala-AI
Python · PyTorch · 2025-2026

- Multi-agent AI framework with four strategy types including a neural network agent trained via self-play reinforcement learning; full training and tournament evaluation pipeline

EXPERIENCE

Python Automation Developer, AllBikes
2022-2023 · Remote

- Built Python automation tools for web scraping and product catalog structuring; **tooling adopted by the full team**, delivered independently in a remote professional environment while in secondary school

EDUCATION

Magshimim Cyber, Advanced Track (Magshimim+)
2023-2026

Israel's elite national cybersecurity program, selective advanced track. Curriculum: networking, binary exploitation, reverse engineering, cryptography, systems programming (C/C++), Linux, team-based security development

BOSMAT, Haifa
2020-2026

Computer Science and AI, 10 study units (maximum track)

AWARDS & RECOGNITION

2nd Place, AI4Y World Competition · 2023 · Global

Magshimim+ Advanced Track · Graduate 2026 **Magshimim Cyber** · Graduate 2026